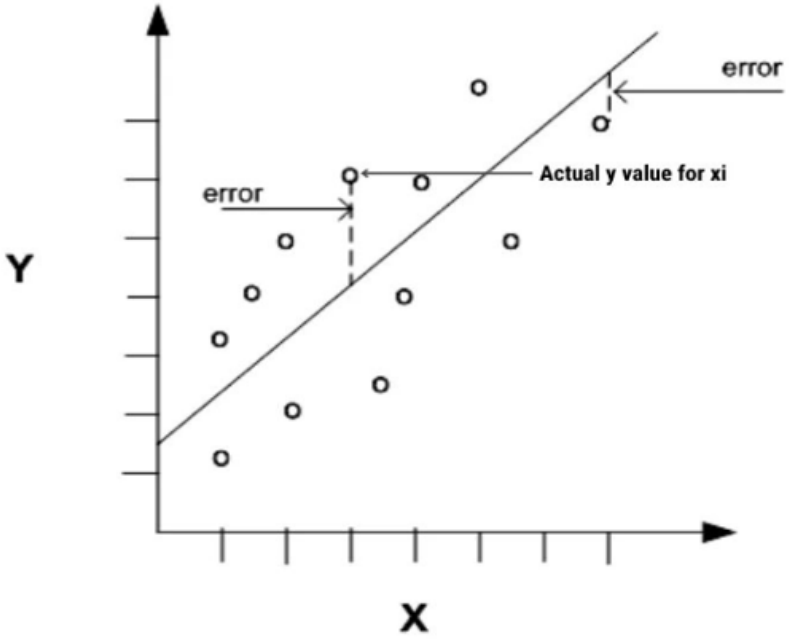
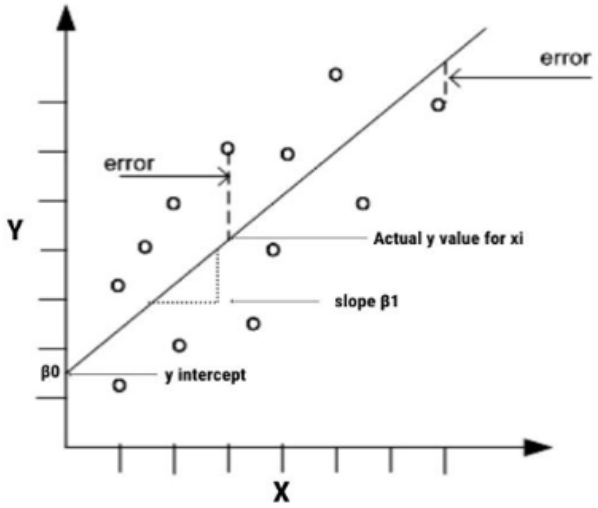
What is Linear regression

Linear regression is a statistical method used to **model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data.** The goal is to find the best-fitting line that minimizes the sum of the squared differences between the observed values and the values predicted by the line.



Equation



The equation for the line is: $Y=\beta0+\beta1X+\varepsilon$ Here,

- Y is the dependent variable,
- X is the independent variable,
- β0 is the y-intercept,
- β1 is the slope of the line, and
- ε represents the error term.

The goal is to find the best-fitting line by adjusting the values of $\beta0$ and $\beta1$ to minimize the sum of squared errors between the observed values and the values predicted by the lin

Code

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

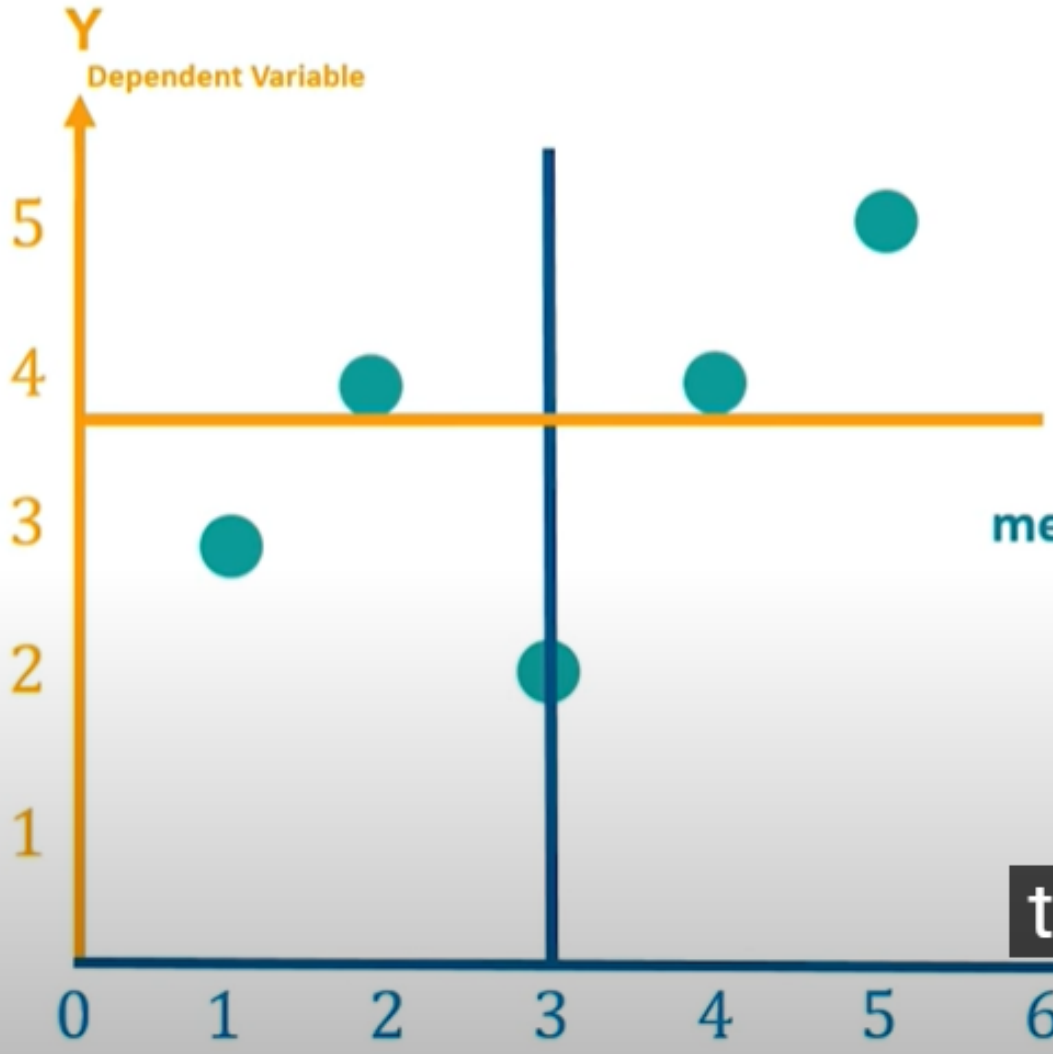
# Generate some random data
np.random.seed(42)
X = 2 * np.random.rand(100, 1)
y = 4 + 3 * X + np.random.randn(100, 1)

# Train a linear regression model
model = LinearRegression()
model.fit(X, y)

# Make predictions
X_new = np.array([[0], [2]])
y_pred = model.predict(X_new)

# Visualize the data and the linear regression line
plt.scatter(X, y, label='Data points')
plt.plot(X_new, y_pred, 'r-',
         label='Linear Regression Line')
plt.xlabel('X')
plt.ylabel('y')
plt.legend()
plt.title('Linear Regression Example')
plt.show()
```

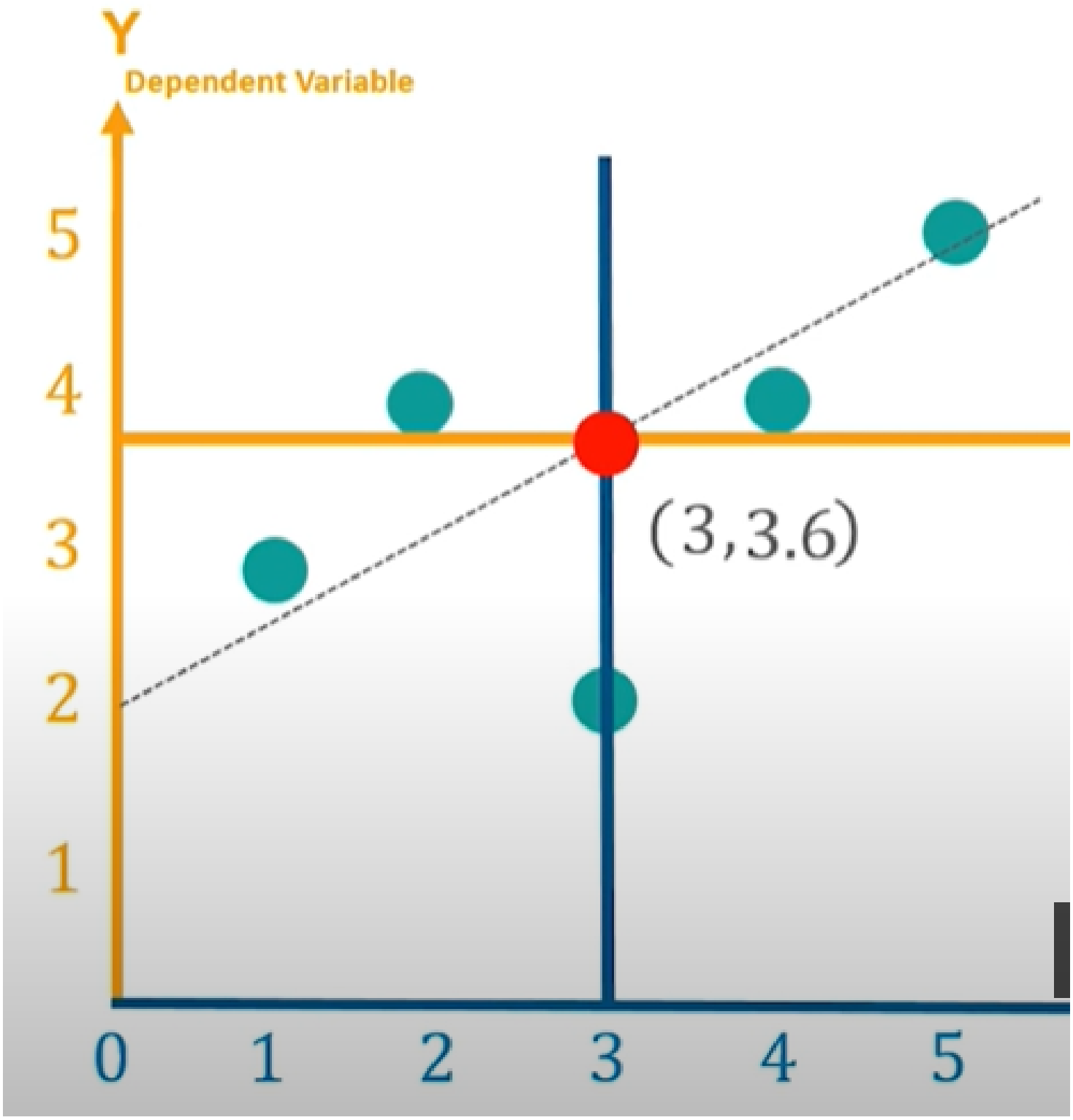
Understanding Linear Regression Al



x	y
1	3
2	4
3	2
4	4
5	5
mean	3	3.6

that is 3 comma 3 .6 on the graph.

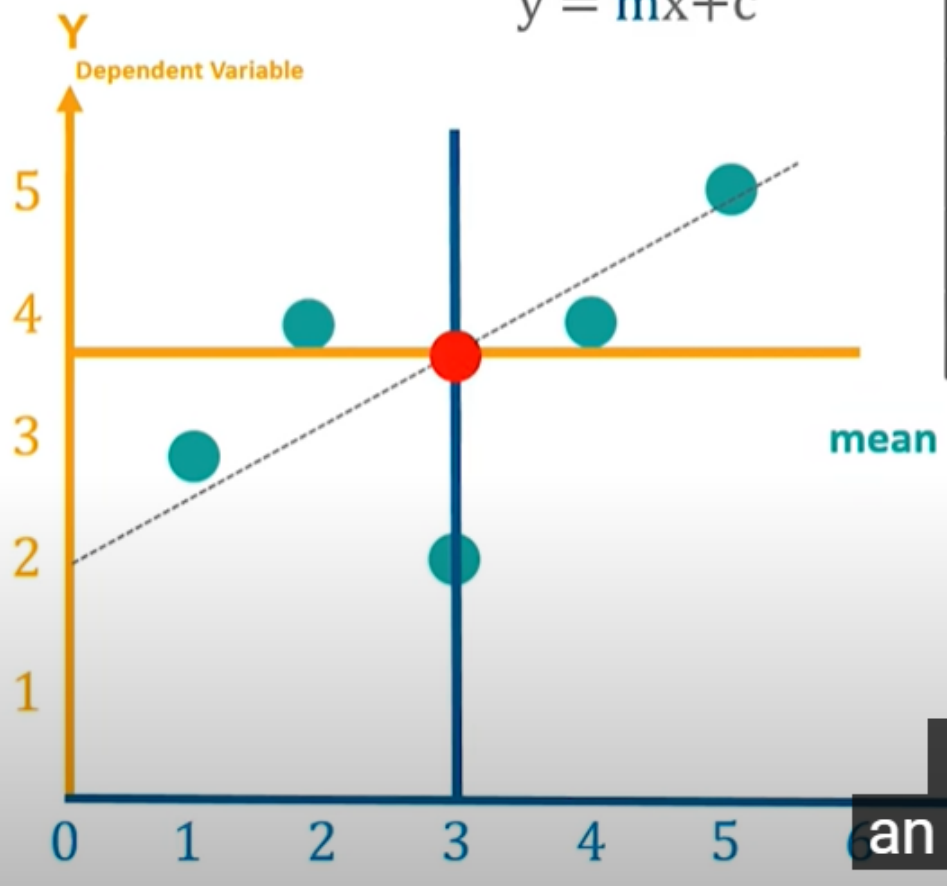
Understanding Linea



Understanding Linear Regression Algorithm

$y = mx+c$

x	y
1	3
2	4
3	2
4	4
5	5

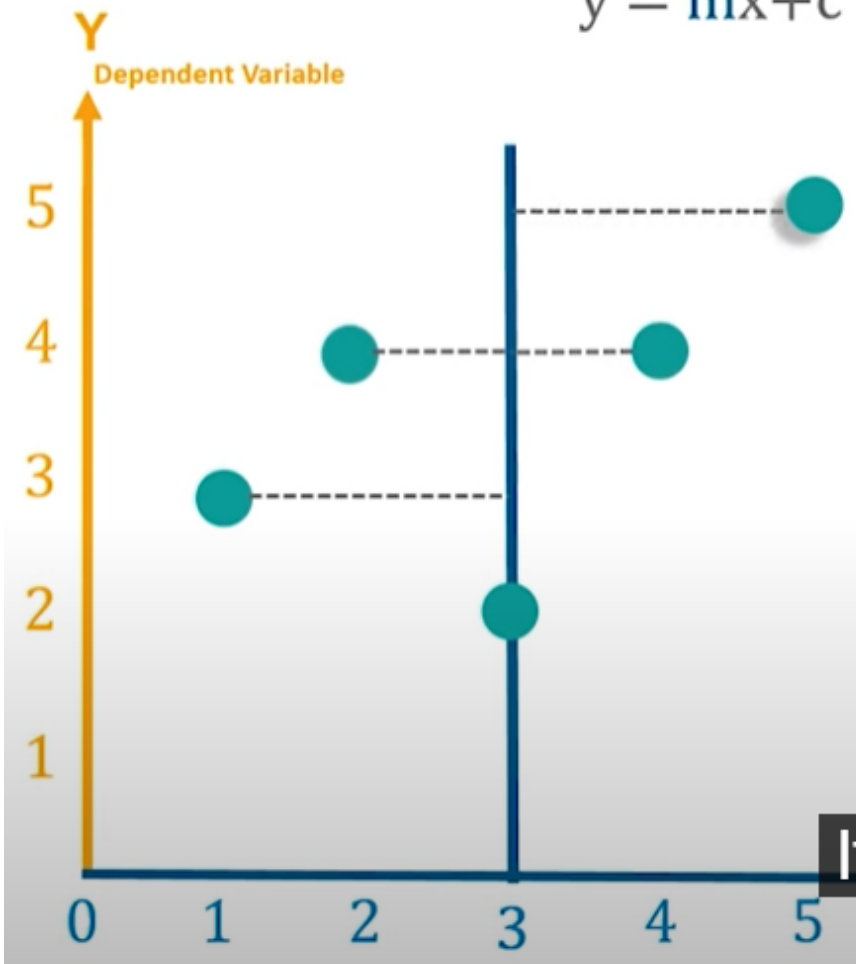


$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Now, we have an equation of line.

Understanding Linear Regression Algorithm

$$y = mx+c$$



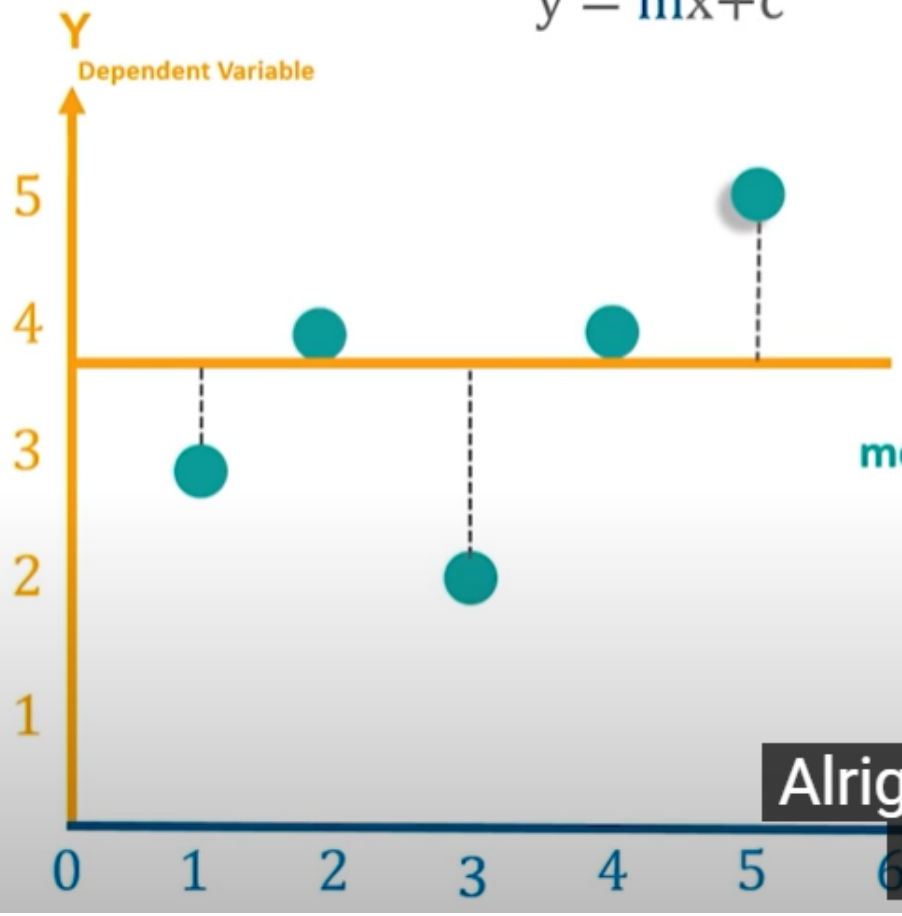
x	y	$x - \bar{x}$
1	3	-2
2	4	-1
3	2	0
4	4	1
5	5	2
mean	3	3.6

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

It's nothing but the distance of all the point

Understanding Linear Regression Algorithm

$y = mx+c$



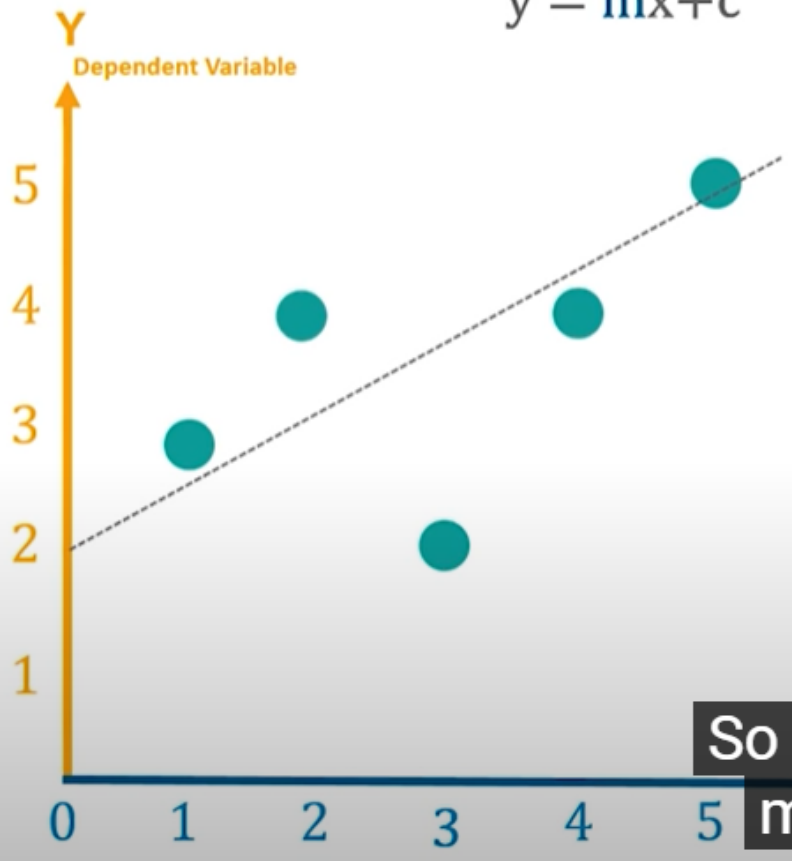
x	y	$x-\bar{x}$	$y-\bar{y}$
1	3	-2	-0.6
2	4	-1	0.4
3	2	0	-1.6
4	4	1	0.4
5	5	2	1.4
mean	3	3.6	

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Alright, so now we are done with Y minus y bar.

Understanding Linear Regression Algorithm

$y = mx+c$



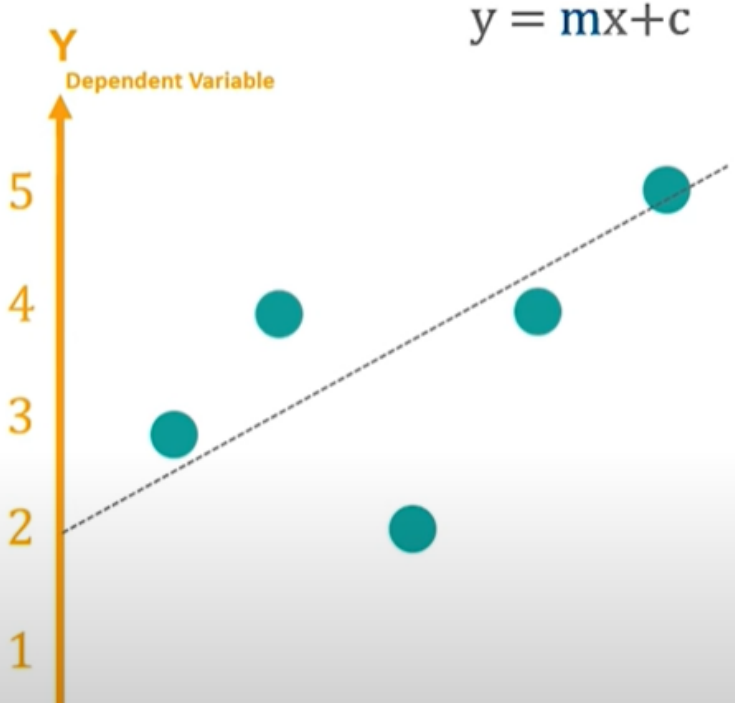
x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$
1	3	-2	-0.6	4
2	4	-1	0.4	1
3	2	0	-1.6	0
4	4	1	0.4	1
5	5	2	1.4	4
mean	3	3.6		

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

So now in our table we have x minus X bar y minus y bar

Understanding Linear Regression Algorithm

$y = mx+c$

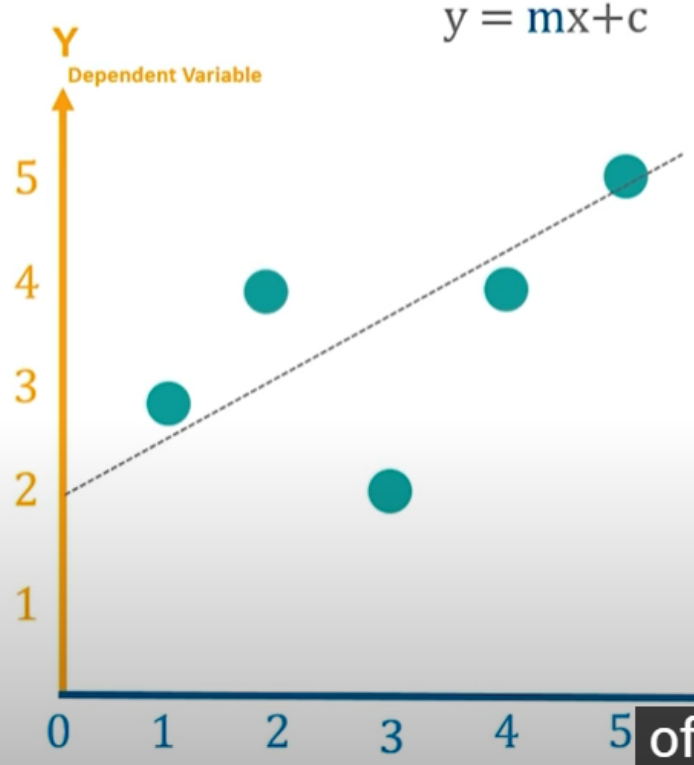


x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
mean	3	3.6			

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression Algorithm

$y = mx+c$



x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 4$

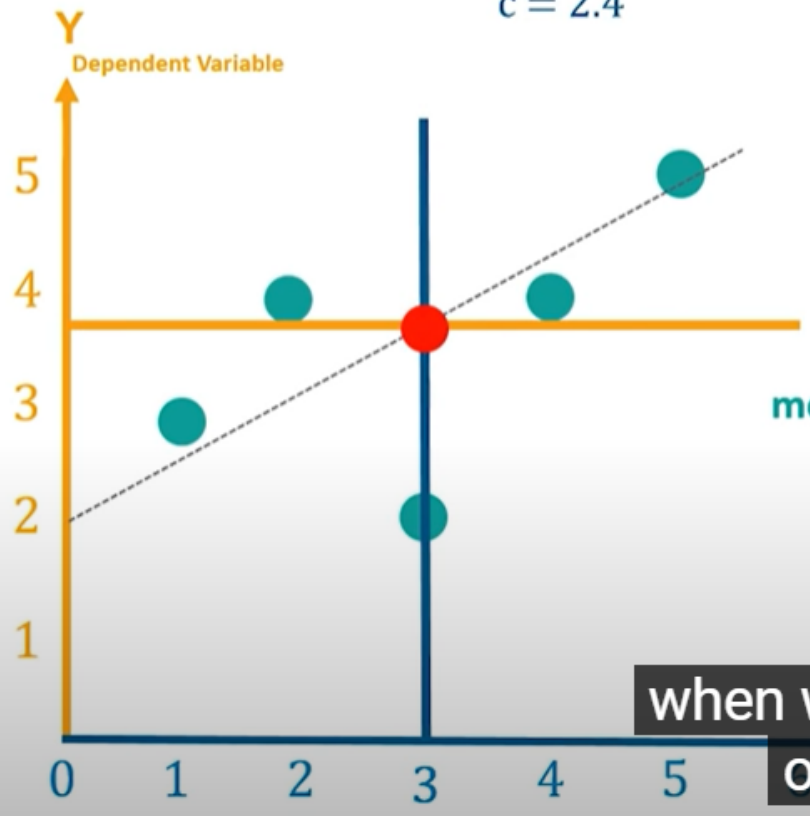
$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2} = \frac{4}{10}$$

So let's put this value
of m equals zero point four

Understanding Linear Regression Algorithm

$y = mx+c$

$c = 2.4$



x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 4$

$$m = \sum \frac{(x - \bar{x})(y - \bar{y})}{(x - \bar{x})^2} = \frac{4}{10}$$

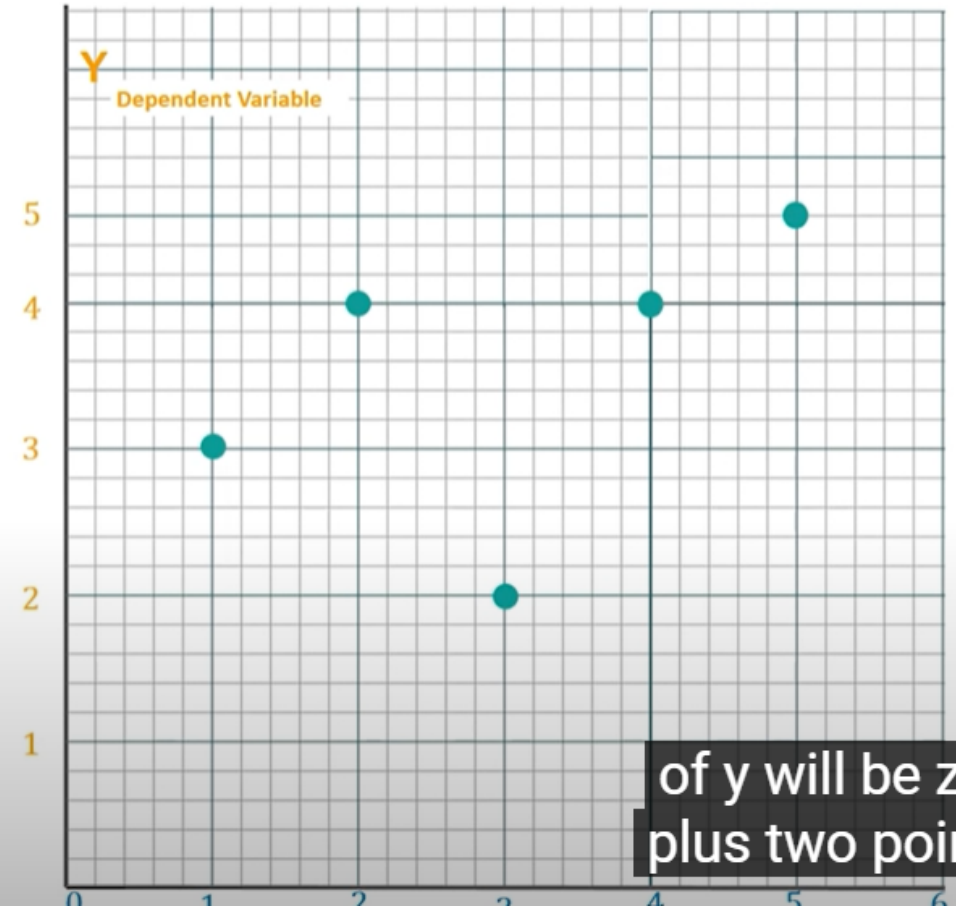
when we calculate the equation of the regression line

$m = 0.4$

$c = 2.4$

$y = 0.4x + 2.4$

Mean Square Error



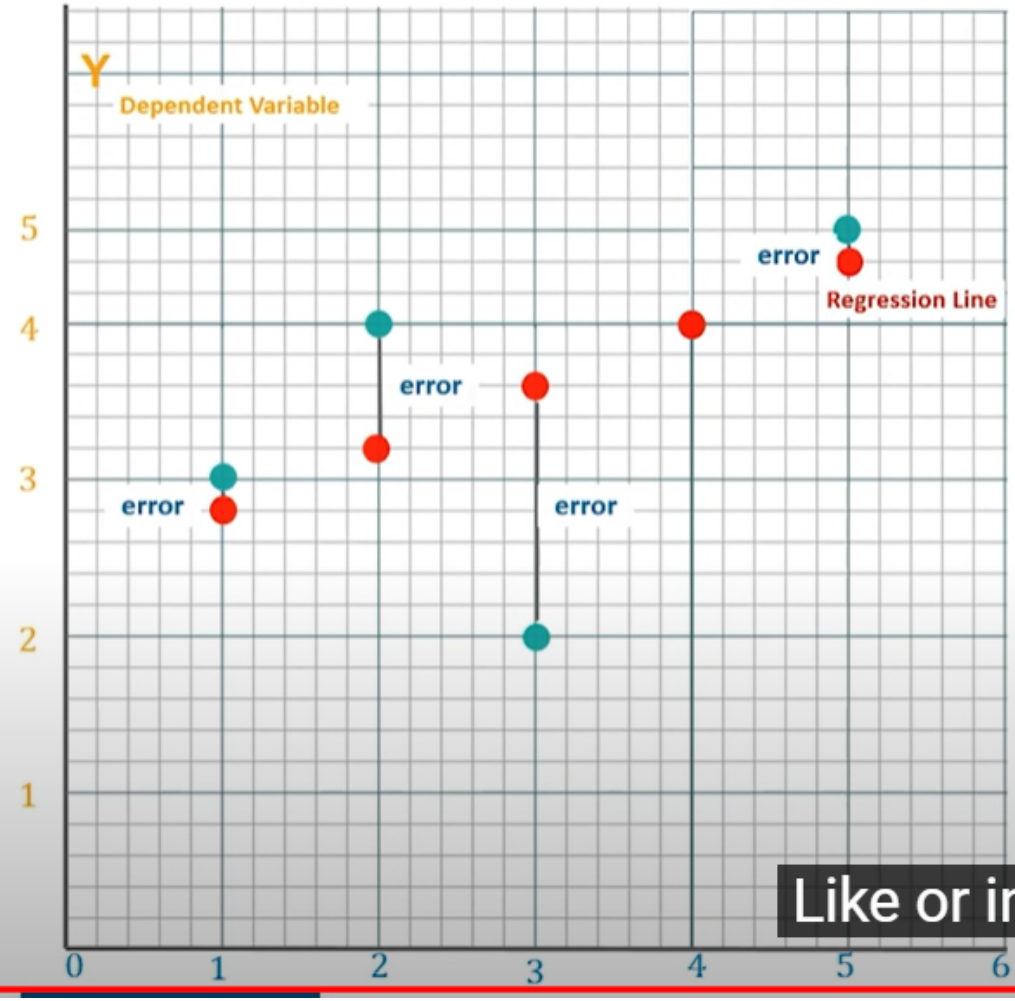
$$m = 0.4$$
$$c = 2.4$$
$$y = 0.4x + 2.4$$

For given $m = 0.4$ & $c = 2.4$, lets predict values for y for $x = \{1,2,3,4,5\}$

$$y = 0.4 \times 1 + 2.4 = 2.8$$
$$y = 0.4 \times 2 + 2.4 = 3.2$$
$$y = 0.4 \times 3 + 2.4 = 3.6$$
$$y = 0.4 \times 4 + 2.4 = 4.0$$
$$y = 0.4 \times 5 + 2.4 = 4.4$$

of y will be zero point 4 x one
plus two point four that is 2.8.

Mean Square Error



Distance between actual
& predicted value

Like or in other words,